SIMATS ENGINEERING

ASSIGNMENT 2 : Problem Solving Assignment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED

CO2: *Analyze and select appropriate agent strategies for different types of environments and problem settings.* (BTL 3)

Total Marks: 100 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Problem Solving Assignment

Code of Conduct:

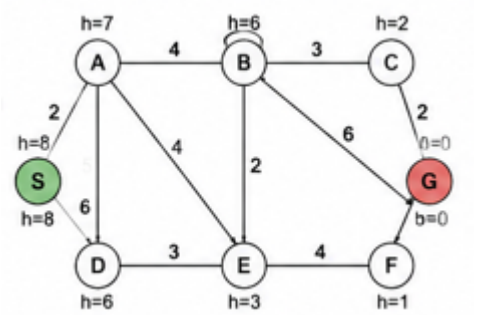
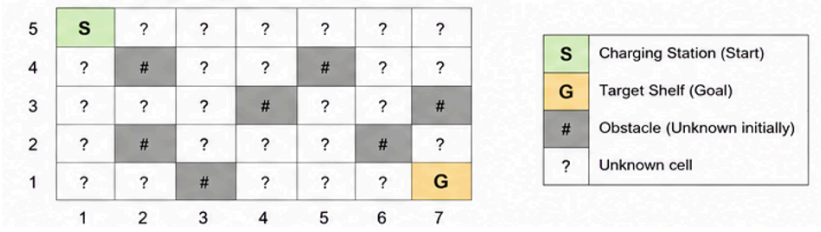
I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Problem Analysis and Understanding	15%	
AI Algorithm Selection & Justification	15%	
Algorithm Design / Pseudocode	25%	
Application of AI Concepts (Greedy, A*, CSP, Minimax, etc.)	15%	
Diagram / State Space / Search Tree Representation	10%	
Complexity Analysis and Solution Efficiency	10%	
Originality, Critical Thinking, and Presentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Answer All the Questions

Q.No	Application-Based Question	Major CO2 Concepts
1	**Smart Ambulance Navigation:** An ambulance must travel from a hospital to an emergency location through a city road network. Some roads have higher travel costs, and an estimated distance to the destination is available for each junction. Design a search-based solution to identify the best route. Using the given road network and heuristic values, determine the route selected by your algorithm, show the sequence of junctions explored, calculate the relevant evaluation values, and justify whether the route is optimal. 	Informed Search, A*, Heuristic Function
2	**Autonomous Warehouse Robot:** A robot has to move from a charging station to a target shelf in a warehouse. The robot does not know all obstacles initially and discovers some obstacles only when it reaches nearby locations. Design an intelligent search strategy for the robot. Demonstrate the robot's decisions for the given grid, explain how it reacts when a new obstacle is discovered, and compare the approach with an offline search strategy. 	Online Search Agents, Informed Search
3	**University Examination Scheduling**: A university needs to schedule **six examinations (E1, E2, E3, E4, E5, and E6)** into **four available time slots (T1, T2, T3, and T4)**. Some examinations have common students and therefore cannot be conducted in the same time slot.	CSP, MRV, Forward Checking, Constraints

The following constraints must be satisfied:

1. E1 and E2 have common students.
2. E1 and E3 have common students.
3. E2 and E4 have common students.
4. E3 and E5 have common students.
5. E4 and E6 have common students.
6. E2 and E5 have common students.

The university has three examination halls with the following capacities:

Examination Hall	Capacity
Hall A	120 students
Hall B	90 students
Hall C	60 students

Examination	E1	E2	E3	E4	E5	E6
Number of Students	100	80	70	60	90	50

Tasks:

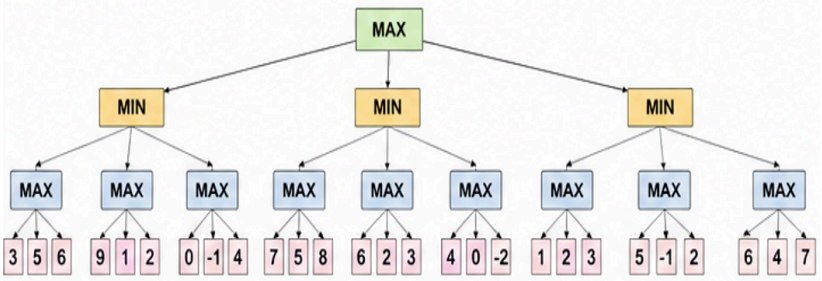
a) Formulate the problem as a **Constraint Satisfaction Problem (CSP)** by clearly defining the **variables, domains, and constraints**.

b) Use **Backtracking with the Minimum Remaining Values (MRV) heuristic and Forward Checking** to obtain a valid examination timetable.

c) Show the **domain of each variable after every assignment** and provide the **final timetable**, ensuring that all student-conflict and hall-capacity constraints are satisfied.

4 **Game-Playing AI:**
You are developing an AI opponent for a board game. The current game position produces the following possible moves and terminal scores in the supplied game tree. Determine the move the AI should make by applying Minimax. Then apply Alpha-Beta Pruning and identify the branches that do not need to be evaluated. Explain how pruning improves the efficiency of the game-playing agent.

Game Playing, Minimax, Alpha-Beta Pruning

		
5	**Delivery Route Optimization:** A logistics company wants to minimize the total distance travelled by a delivery vehicle visiting multiple customers. An initial route and several neighboring routes are provided. Design an optimization strategy to improve the route. Apply the strategy step-by-step to the given routes, determine the final route, explain whether it represents a global optimum, and suggest how the algorithm could escape a local optimum. **Distance matrix** (in km) | | D | A | B | C | E | |---|---|---|---|---|---| | D | – | 6 | 9 | 8 | 7 | | A | 6 | – | 3 | 5 | 6 | | B | 9 | 3 | – | 4 | 5 | | C | 8 | 5 | 4 | – | 3 | | E | 7 | 6 | 5 | 3 | – | **Some neighboring routes of the initial solution and their costs** S1: D → A → C → B → E → D = 26 km S2: D → B → A → C → E → D = 30 km S3: D → A → B → E → C → D = 24 km S4: D → E → C → B → A → D = 27 km S5: D → A → E → B → C → D = 23 km S6: D → C → B → A → E → D = 29 km From S5, the best neighbor has cost 22 km. From that solution, all neighbors have cost ≥ 22 km. Apply the chosen local search strategy and report the final route.	Local Search, Optimization, Hill Climbing

EVALUATION RUBRICS

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
CSP Problem Formulation	20%	4% – Clearly defines all variables, domains, and problem objectives correctly.	3% – Defines most variables and domains with minor errors.	2% – Partial formulation with some missing components.	0–1% – Incorrect or incomplete formulation.
Constraint Identification & Representation	20%	4% – Accurately identifies all student-conflict and hall-capacity constraints and represents them correctly.	3% – Identifies most constraints with minor errors.	2% – Identifies some constraints but misses important relationships.	0–1% – Constraints are largely incorrect or missing.
Application of Backtracking, MRV & Forward Checking	30%	6% – Correctly applies Backtracking, MRV, and Forward Checking with clear and logical steps.	4.5% – Applies the techniques correctly with minor mistakes.	3% – Partial application with unclear or inconsistent steps.	0–1.5% – Incorrect or inappropriate application of CSP techniques.
Step-by-Step Execution & Domain Updates	20%	4% – Provides a complete, logical execution trace and correctly updates domains after every assignment.	3% – Mostly correct execution with minor gaps.	2% – Provides limited steps with some incorrect domain updates.	0–1% – Incomplete or incorrect execution.
Final Timetable, Accuracy & Justification	10%	2% – Produces a valid conflict-free timetable satisfying all constraints and capacity requirements with clear justification.	1.5% – Valid timetable with minor errors in justification.	1% – Partially valid timetable with some constraint violations.	0–0.5% – Invalid or missing timetable.